Multifeature Collaborative Adversarial Attack in Multimodal Remote Sensing Image Classification

Cheng Shi, *Member, IEEE*, Yenan Dang, Li Fang, *Member, IEEE*, Minghua Zhao, Zhiyong Lv, *Member, IEEE*, Qiguang Miao, *Senior Member, IEEE*, and Chi-Man Pun, *Senior Member, IEEE*

***Abstract*— Deep neural networks have strong feature learning ability, but their vulnerability cannot be ignored. Current research shows that deep learning models are threatened by adversarial examples in remote sensing (RS) classification tasks, and their robustness drops sharply in the face of adversarial attacks. Therefore, many adversarial attack methods have been studied to predict the risks faced by a network. However, the existing adversarial attack methods mainly focus on single-modal image classification networks, and the rapid growth of RS data makes multimodal RS image classification a research hotspot. Generating multimodal adversarial examples needs to consider a high attack success rate, subtle perturbation, and collaborative attack ability between different modalities. In this article, we investigate the vulnerability of multimodal RS classification networks and propose a multifeature collaborative adversarial network (MFCANet) for generating multimodal adversarial examples. Two modality-specific generators are designed to generate the multimodal collaborative perturbations with strong attack ability, and two modality-specific discriminators make the generated multimodal adversarial examples closer to the real instances. In addition, a modality-specific generative loss and a modality-specific discriminative loss are proposed, and an alternating optimization strategy is designed for training the proposed MFCANet. Extensive experiments are carried out on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen 2D dataset and ISPRS Potsdam 2D dataset. The results show that the attack performance of the proposed method is stronger than that of the fast gradient sign method (FGSM), project gradient descent (PGD), and Carlini and Wagner (C&W) attack methods.**

***Index Terms*— Generative adversarial networks (GANs), multimodal adversarial attack, multimodal remote sensing (RS) image classification.**

Manuscript received 17 May 2022; revised 14 August 2022 and 7 September 2022; accepted 15 September 2022. Date of publication 21 September 2022; date of current version 24 October 2022. This work was supported by the National Natural Science Foundation of China under Grant 61902313, Grant 61973250, Grant 42101359, Grant 62002272, and Grant 61902296. *(Corresponding author: Li Fang.)*

Cheng Shi, Yenan Dang, Minghua Zhao, and Zhiyong Lv are with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, Shaanxi 710048, China (e-mail: chengc_s@163.com; nn1262963445@163.com; mh_zhao@126.com; lvzhiyong_fly@hotmail.com).

Li Fang is with the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou, Fujian 362216, China (e-mail: fangli@fjirsm.ac.cn).

Qiguang Miao is with the School of Computer Science and Technology, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: qgmiao@xidian.edu.cn).

Chi-Man Pun is with the Department of Computer and Information Science, University of Macau, Macau, China (e-mail: cmpun@umac.mo).

Digital Object Identifier 10.1109/TGRS.2022.3208337

I. Introduction

REMOTE sensing (RS) image classification (semantic segmentation) aims to assign a semantic label for each pixel, which is still a challenging task in the RS domain [1], [2], [3]. Multimodal RS data have different observations for land cover, such as true orthophoto (TOP) data proving the high spatial-resolution information of the sense and digital surface model (DSM) data proving the height information of the objects [4]. Based on the advantages of multimodal RS data, multimodal classification methods have been studied to reduce the uncertainty of classification [5]. In recent years, deep neural networks have been widely introduced into multimodal RS image classification tasks [6], [7], [8]. The basic idea of multimodal RS image classification is to learn modality-specific discriminative features with multilayered deep architecture, and the features of different streams are fed into a joint representation. Compared with single-modal RS classification methods, multimodal RS classification methods can achieve higher classification accuracy by exploiting the correlation between the two modalities.

Nevertheless, the deep learning model has superior feature learning ability, but unfortunately its vulnerability was reported by Szegedy *et al.* [9]. They found that inserting subtle perturbations into a legitimate image caused the networks to make incorrect predictions. Examples that make the networks misclassify are known as adversarial examples [10]. Then, several works studied the vulnerability of the deep learning model in many different tasks, especially in the natural image classification task [11], [12]. The vulnerability of deep learning models has also attracted the attention of RS researchers. In the RS scene classification task [13], [14], SAR image classification task [15], and hyperspectral image classification task [16], a deep learning model also shows strong sensitivity in the face of adversarial examples, and the robustness of the networks is greatly reduced.

We need to know the risks faced by the networks in advance, which is beneficial for evaluating and improving the robustness of a deep learning model. Therefore, adversarial attack methods were studied to find the subtle perturbation for generating adversarial examples [17], [18], [19]. These adversarial attack methods have proven the effectiveness of the attack, which can drastically reduce the performance

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

of the attacked model (target model). However, the current adversarial attack methods mainly focus on attacking single-modal classification networks and lack research on adversarial attacks on multimodal deep learning models.

To analyze the robustness of multimodal RS classification networks, we propose an adversarial attack method for attacking multimodal RS classification networks, named multifeature collaborative adversarial networks (MFCANets). Compared with the single-modal adversarial attack method, the perturbations are added to both data sources in the multimodal adversarial attack. Therefore, two aspects need to be considered to solve the multimodal perturbations: 1) the generated multimodal adversarial examples should be closer to real multimodal instances and 2) the perturbations on multiple data sources should have collaborative attack abilities. We apply generative adversarial networks (GANs) to produce multimodal perturbations. The RS images usually contain more complex scene information, and directly generating complex adversarial examples leads to the instability of GAN training. Therefore, the modality-specific generator is designed to produce the perturbations added to the legitimate examples, and the modality-specific discriminator is constructed to restrict the generated adversarial examples to be close to the real instances. In addition, a collaborative attack loss is designed to capture the relationship between the perturbations of different modalities. The collaborative attack can effectively reduce the intensity of the perturbations added to each data source while maintaining a higher attack success rate.

To evaluate the effectiveness of the proposed MFCANet method, multimodal adversarial examples are generated with three state-of-the-art attack methods. The quality of the generated multimodal adversarial examples is evaluated in two aspects: one is the attack success rate, and the other is the distance between the adversarial examples and classification boundary. The distance is difficult to measure numerically; therefore, a defensive strategy with adversarial training is used to evaluate the attack performance. The classification results on the original target model and defensive model show that the multimodal adversarial examples generated by MFCANet have more superior attack performance. This superiority is possibly due to the adversarial examples of the two modalities being closer to their real instances and having stronger collaborative attack capabilities.

The main contributions of this study are listed as follows.

1) We propose a novel multimodal collaborative adversarial attack method for attacking multimodal RS classification networks. The proposed multimodal adversarial attack method trains multimodal adversarial perturbations with modality-specific GANs, which can obtain adversarial examples that are closer to the classification boundary. To the best of our knowledge, this is the first study on attacking multimodal RS classification networks.
2) We develop a collaborative attack loss to improve the collaborative attack ability among the perturbations of different modalities. The collaborative attack loss can achieve stronger attack performance with fewer perturbations.
3) An iterative optimization strategy is designed for training the modality-specific generator and modality-specific discriminator to improve the optimization efficiency.
4) Extensive experiments are conducted on two open International Society for Photogrammetry and Remote Sensing (ISPRS) benchmark datasets, and the dates of TOP and DSM are used to perform a multimodal classification task. We test the attack success rate of legitimate examples and adversarial examples on the target and defensive models, respectively, and show that MFCANet achieves stronger attack performance and that the generated adversarial examples are closer to the classification boundary.

The rest of this article is organized as follows. A literature review is given in Section II. Section III elaborates on the proposed method. The experimental results are presented in Section IV. A discussion is given in Section V, and the conclusions and future work are discussed in Section VI.

II. Literature Review

A. State-of-the-Art Adversarial Attack Methods

The adversarial examples can make a well-trained model output error prediction. In this section, several state-of-the-art adversarial attack methods used in our experiments are briefly reviewed.

1) Fast Gradient Sign Method (FGSM): FGSM was proposed by Goodfellow *et al.* [17] and is a one-step method to generate adversarial examples. The adversarial examples are generated according to the following equation:

$$\begin{aligned} x' &= x + \varepsilon \operatorname{sgn}(\nabla_x L(x, y)), \quad \text{non-target} \\ x' &= x - \varepsilon \operatorname{sgn}(\nabla_x L(x, t)), \quad \text{target on } t \end{aligned} \tag{1}$$

where x is a legitimate image, and y is its true label. $\operatorname{sgn}(\cdot)$ is the sign function. $\nabla_x L(x, y)$ denotes the gradient of the objective function $L(x, y)$ to the input legitimate image x. Equation (1) can be considered a solution to the optimization problems, as shown in the following equation (taking the target attack as an example):

$$\min L(x, t), \quad \text{s.t. } \|x' - x\|_\infty \leqslant \varepsilon. \tag{2}$$

Equation (2) searches an adversarial example x' to classify t in an ε-neighbor ball of legitimate image x. The generated adversarial examples can fool the deep learning model with small perturbations. FGSM involves calculating the gradient once and has a faster stack speed.

2) Project Gradient Descent (PGD): PGD [18] is an iterative version of the one-step FGSM. The adversarial examples produced by PGD are shown in the following equation (target attack):

$$x_{i+1} = \operatorname{clip}_{x,\varepsilon}(x_i + \mu \operatorname{sgn}(\nabla_{x_i} L(x_i, t)) \tag{3}$$

where $x_0 = x$, and μ is the iterative step. $\operatorname{clip}(\cdot)$ is for constraining x_i within the ε-neighbor ball of legitimate image x.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

3) Carlini and Wagner (C&W): The C&W attack algorithm [19] was proposed by Carlini and Wanger and solves perturbations based on optimization, as shown in the following equation:

$$\min \|x - x'\|_2^2 + cf(x', t). \tag{4}$$

The authors proposed six candidates for the function $f(\cdot)$. In this study, $f(\cdot)$ is selected as $f(x', t) = (\max_{i \neq t} Z(x')_i - Z(x')_t)^+$. c is a constant.

4) Generative Adversarial Network (GAN)-Based Adversarial Attack Methods: Compared to gradient-based and optimization-based adversarial attack methods, GAN-based methods generate adversarial examples that are closer to real instances [12], [20], [21]. Xiao *et al.* [22] proposed an AdvGAN method to improve the perceptual quality of the generated adversarial examples, and the adversarial examples were generated with GANs. Jandial *et al.* [23] proposed an advGAN++ method in which the latent features serve as prior knowledge for adversarial generation. Liu and Hsieh [24] proposed a Rob-GAN method in which the generator and adversarial attacker can benefit each other in the training process. Cheng *et al.* [25] proposed a perturbation-seeking GAN model to generate more adaptive adversarial examples. Different from the aforementioned method, we focus on generating collaborative perturbations in multimodal RS images based on GANs.

B. Adversarial Attack in the Field of RS

The vulnerability of natural image classification networks has also attracted the attention of researchers in the field of RS image analysis. Xu *et al.* [13] and Chen *et al.* [14] investigated the influence of adversarial examples on RS scene classification tasks. They found that attack algorithms can fool several state-of-the-art RS scene classification networks, and the robustness of the networks was drastically reduced. Li *et al.* [15] studied the robustness of SAR image classification networks and analyzed the vulnerability and sensitivity of the networks in the face of adversarial examples. Xu *et al.* [16] proved the existence of adversarial examples in hyperspectral image classification networks and presented a defensive strategy to defend against hyperspectral adversarial examples. Du *et al.* [26] proposed a fast-C&W adversarial attack algorithm to reduce attack time for SAR target recognition tasks. In our previous work, we proposed a targetfool adversarial attack algorithm for hyperspectral image classification networks [27]. Although the aforementioned works have carried out preliminary explorations of adversarial attacks in the field of RS, the study is still in its infancy, especially regarding on adversarial examples in multimodal RS classification tasks.

C. Adversarial Training

Adversarial training is an effective defense strategy against adversarial examples. In adversarial training, the generated adversarial examples are incorporated into the training set to retrain the target model [10], [28]. This defense strategy is simple and effective, but the defense effect depends on the quality of the adversarial examples. The retrained defensive model should resist the attacks of adversarial examples while reducing the impact on the classification accuracy of legitimate examples. In this article, an adversarial training strategy is used to evaluate the attack performance of the proposed attack model.

III. Proposed Method

Fig. 1 shows the architecture of the proposed MFCANet. Assume that the two input modalities are TOP (R-G-B three bands) and DSM (one band). The architecture of MFCANet consists of three parts: two modality-specific generators G_T and G_D, two modality-specific discriminators D_T and D_D, and a target model f. The modal-specific generators are designed to generate the perturbations of each modality. The modality-specific discriminators are designed to distinguish the legitimate examples and generated adversarial examples. The target model encourages the generated examples to be misclassified. The proposed modal has an end-to-end form. The details of the proposed method are described in the following.

A. Multifeature Collaborative Adversarial Attack

1) Modality-Specific Generator: The RS images have complex scene information; hence, two generators are constructed to generate the perturbation of each modality. Assume that the two input modalities are x_T and x_D. The modality-specific generator G_T takes the modality x_T as its input, and the modality-specific generator G_D takes the modality x_D as its input. The output of each generator is the perturbation of the adversarial example, and the outputs are denoted as $G_T(x_T)$ and $G_D(x_D)$ for each modality, respectively. The generated adversarial examples can be obtained by adding the perturbations on the input legitimate examples, which are denoted as $x'_T = x_T + G_T(x_T)$ and $x'_D = x_D + G_D(x_D)$, respectively.

To optimize the modality-specific generator, a modality-specific generative loss is designed, including adversarial loss, perceptual loss, fooling loss, and collaborative attack loss.

The adversarial loss is defined in the following equations:

$$L_{\text{adv}}^{G_T} = E_{x_T} \log D_T\left(x'_T\right) \tag{5}$$

$$L_{\text{adv}}^{G_D} = E_{x_D} \log D_D\left(x'_D\right). \tag{6}$$

The inputs of the modality-specific discriminators D_T and D_D are the generated adversarial examples, respectively, aiming at encouraging the discriminator to discriminate the generated adversarial examples as the real class.

The perceptual loss is designed to constrain the strength of the perturbation and is shown in the following equations:

$$L_p^{G_T} = E_{x_T}\{\max(\varepsilon, \|G_T(x_T)\|_p)\} \tag{7}$$

$$L_p^{G_D} = E_{x_D}\{\max(\varepsilon, \|G_D(x_D)\|_p)\} \tag{8}$$

where $L_p^{G_T}$ and $L_p^{G_D}$ are the modality-specific perceptual losses. ε is a hyperparameter that controls the minimum allowable perturbation strength. $\|\cdot\|_p$ denotes the l_p norm of the perturbation. The purpose of perceptual loss is to force

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

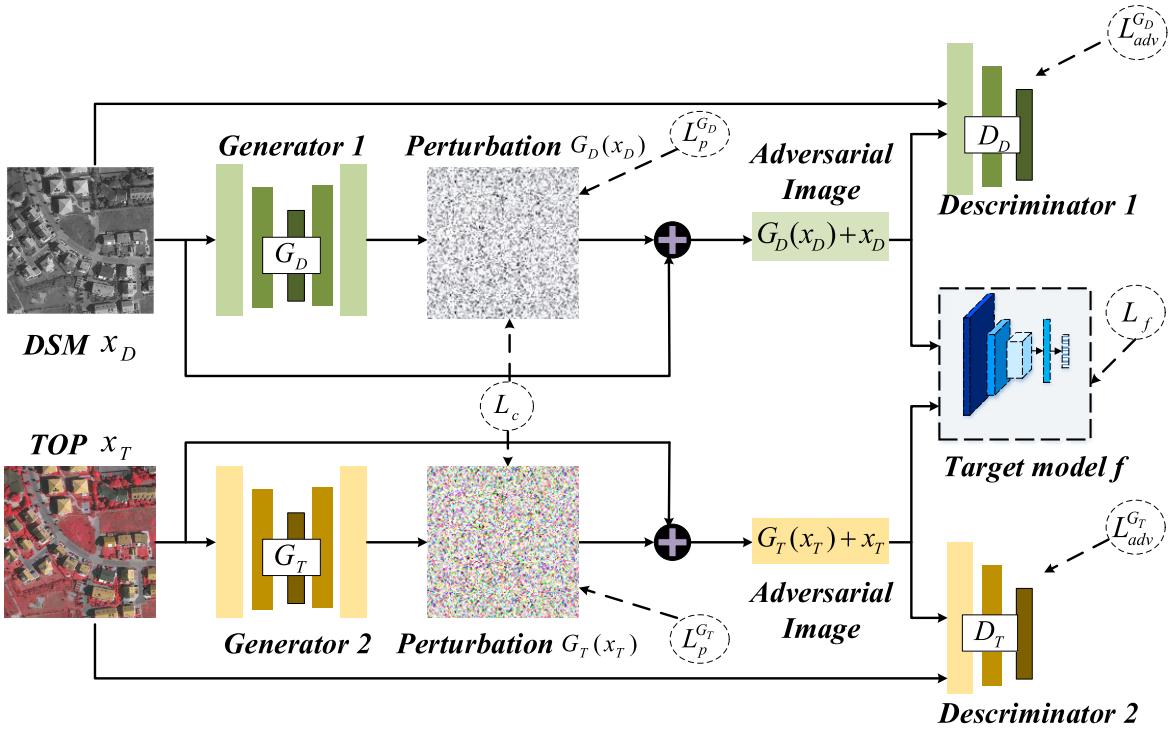


Fig. 1. Architecture of MFCANets.

the generated adversarial examples closer to the real instance. The perceptual loss and adversarial loss have a collaborative inhibitory effect on perturbation generation, which can reduce the perceptual distance between the legitimate examples and adversarial examples. In the following, we use the l_∞ norm to constrain the perturbation.

The fooling loss encourages the predictions of the multimodal classification networks to be misclassified into the target class t, as shown in the following equation:

$$L_f = l_f\left(f\left(x'_T, x'_D\right), t\right) \tag{9}$$

where f is a well-trained multimodal classification network that accepts two different modalities as inputs. l_f is the cross-entropy loss. For an untargeted attack, the fooling loss is defined in the following equation:

$$L_f = -l_f\left(f\left(x'_T, x'_D\right), y\right) \tag{10}$$

where y is the true label. Equation (10) maximizes the distance between the predictions of the multimodal classification networks and the true label. In the following, we mainly focus on the targeted attack, and the target label t is the location of the second-largest probability in the output predictions.

The collaborative attack loss is designed to achieve a multimodal collaborative attack and is defined in the following equation:

$$L_c = l_c(T(G_T(x_T)), G_D(x_D)) \tag{11}$$

where $G_T(x_T)$ and $G_D(x_D)$ are the generated perturbations of each modality. $l_c(\cdot)$ is a similar metric function. In this article, the cosine similarity metric is used to measure the similarity of perturbations in different modalities. The TOP data and DSM data have a different number of bands, and the perturbation of TOP has three bands, while the perturbation of DSM has one band; therefore, we need to expand the perturbation of the DSM into three bands for similarity calculation, and $T(\cdot)$ is the band transform function, which we implement by simply repeating the DSM band. The lack of collaborative attack loss may fall into two situations: 1) the perturbation on one of the modalities plays a leading role in attacking the target model, which will cause a large perturbation added on a single modality and 2) a large perturbation is added on both modalities to achieve a stronger attack. Due to the limitation of the allowable perturbation strength, the added perturbation cannot achieve the optimal attack effect. The collaborative attack loss can constrain the perturbations between different modalities to have similar distributions, and find the optimal perturbation combination under this constraint. Further explanation is verified experimentally in Section V-B.

The full loss for generating the modality-specific adversarial perturbation is shown in the following equation:

$$L_G = L_{\text{adv}}^{G_T} + L_{\text{adv}}^{G_D} + \alpha\left(L_p^{G_T} + L_p^{G_D}\right) + \beta L_f + \gamma L_c \tag{12}$$

where α, β, and γ are the weight coefficients of the perceptual loss term, fooling loss term, and collaborative loss term, respectively.

2) Modality-Specific Discriminator: The discriminator is a binary classifier that aims to discriminate between the generated adversarial examples and legitimate examples. The discriminator loss of TOP and DSM is defined in (13)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

and (14), respectively,

$$L_{\text{adv}}^{D_T} = E_{x_T} \log D_T(x_T) + E_{x_T} \log\left(1 - D_T\left(x_T'\right)\right) \quad (13)$$

$$L_{\text{adv}}^{D_D} = E_{x_D} \log D_D(x_D) + E_{x_D} \log\left(1 - D_D\left(x_D'\right)\right). \quad (14)$$

In (13) and (14), the legitimate examples x_T and x_D have true class, and the labels of the generated adversarial examples x_T' and x_D' are false. The full objective loss function is shown in the following equation:

$$L_D = L_{\text{adv}}^{D_T} + L_{\text{adv}}^{D_D}. \quad (15)$$

B. Model Training With Iterative Optimization Strategy

An alternative optimization approach is designed for optimizing the proposed modality-specific generative loss and modality-specific discriminative loss.

1) Update TOP Discriminator D_T: By fixing the DSM discriminator D_D, TOP generator G_T, and DSM generator G_D, the adversarial examples of the TOP are predicted via a forward calculation. Feed the predicted TOP adversarial examples and the corresponding legitimate examples into the TOP discriminator. Update the discriminator D_T according to the following equation:

$$\arg\min_{D_T} L_{D_T} = \arg\min_{D_T} E_{x_T} \log D_T(x_T) + E_{x_T} \log\left(1 - D_T\left(x_T'\right)\right). \quad (16)$$

2) Update TOP Generator G_T: The TOP discriminator D_T, DSM discriminator D_D, and DSM generator G_D are fixed, the legitimate examples of TOP and DSM are input into the proposed attack model, and the TOP generator is updated according to the update formula, as shown in the following equation:

$$\arg\min_{G_T} L_{G_T} = \arg\min_{G_T}\left(L_{\text{adv}}^{G_T} + \alpha L_p^{G_T} + \beta L_f + \gamma L_c\right). \quad (17)$$

3) Update DSM Discriminator D_D: The TOP discriminator D_T, TOP generator G_T, and DSM generator G_D are fixed according to the generated DSM adversarial examples and corresponding legitimate examples to update the DSM discriminator, as shown in the following equation:

$$\arg\min_{D_D} L_{D_D} = \arg\min_{D_D} E_{x_D} \log D_D(x_D) + E_{x_D} \log\left(1 - D_D\left(x_D'\right)\right). \quad (18)$$

4) Update DSM Generator G_D: The TOP discriminator D_T, DSM discriminator D_D, and TOP generator G_T are fixed, taking the legitimate examples of TOP and DSM as inputs, according to the prediction results, to update the DSM generator, as shown in the following equation:

$$\arg\min_{G_D} L_{G_D} = \arg\min_{G_D}\left(L_{\text{adv}}^{G_D} + \alpha L_p^{G_D} + \beta L_f + \gamma L_c\right). \quad (19)$$

IV. Experiments

A. Dataset Description

1) ISPRS Vaihingen 2D Dataset: The ISPRS Vaihingen 2D dataset[1] is provided by the ISPRS. The dataset consists of a very high-resolution TOP tile and corresponding DSM.

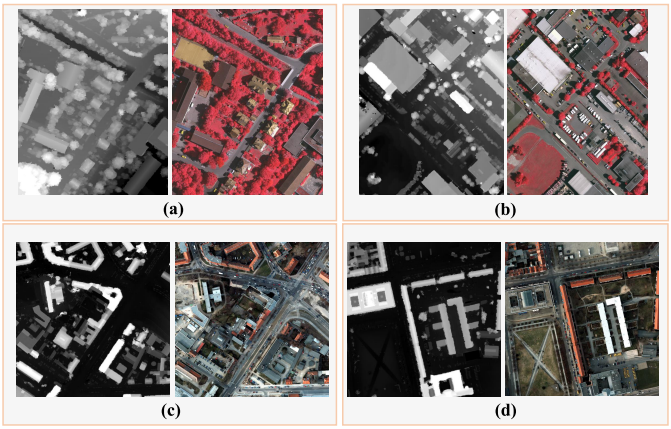


Fig. 2. ISPRS Vaihingen 2D dataset with TOP and DSM. (a) Vaihingen_20 with an image size of 1866 × 2315. (b) Vaihingen_31 with an image size of 1980 × 2555. The ISPRS Potsdam 2D dataset with TOP and DSM. (c) Potsdam5_15 with an image size of 6000 × 6000. (d) Potsdam6_13 with an image size of 6000 × 6000.

Six categories are contained in the dataset: impervious surfaces, building, low vegetation, tree, car, and clutter/background. There are 33 patches contained in the ISPRS Vaihingen dataset, and Fig. 2(a) and (b) shows the DSM and TOP with patch ID numbers "Vaihingen_20" and "Vaihingen_31."

2) ISPRS Potsdam 2D Dataset: The ISPRS Potsdam 2D dataset[2] is provided by the ISPRS. The dataset contains six categories, including impervious surfaces, building, low vegetation, tree, car, and clutter/background. There are 38 patches contained in the ISPRS Potsdam dataset, and each patch consists of a TOP tile and a DSM. Fig. 2(c) and (d) shows the DSM and TOP with patch ID numbers "Potsdam5_15" and "Potsdam6_13."

Since the target model is a pixel-level classification model, each patch is used as independent experimental data. For each patch, we randomly select some pixels to construct training examples for training the attack model, and the remaining pixels are used for testing.

B. Experimental Setting

The parameters of the modality-specific generator, modality-specific discriminator, and target model are listed in Fig. 3. For the ISPRS Vaihingen 2D dataset, 800 pixels per class are randomly selected for training, and the remaining pixels are used for testing. For the ISPRS Potsdam 2D dataset, because of the larger image size, 10 000 pixels per class are randomly selected for training, and the remaining pixels are selected for testing. For each pixel, we choose 28 × 28 local patches as the input of the model. The hyperparameters α, β, and γ in (12) are set as 0.5, 10, and 0.7, respectively, and the hyperparameter ε in perceptual loss [see (7) and (8)] is set as 0.3. The hyperparameters are discussed in Section V.

In the following experiments, the OA, kappa, and mIoU values of legitimate examples and adversarial examples are tested on the target model and the defensive model, respectively. The numerical metric on the target model analyzes the

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CONDITIONAL EXPLANATION OF FIG. 4

Model	Target model	Defensive model	
Fig.4	Adversarial examples are close to the classification boundary	Classification accuracy of legitimate examples	Classification accuracy of adversarial examples
(a)	Yes	High	High
(b)	Part close/part away	High	Low
(c)	No	Low	High

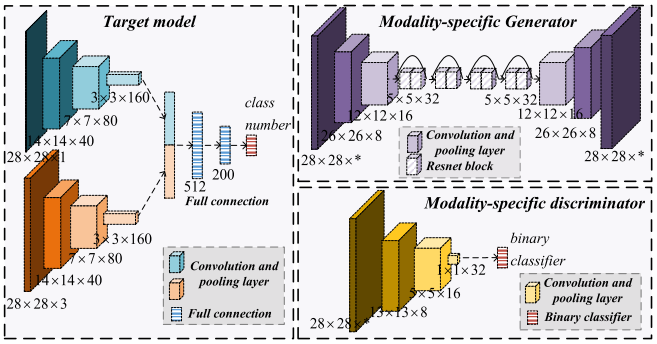


Fig. 3. Parameters of the model.

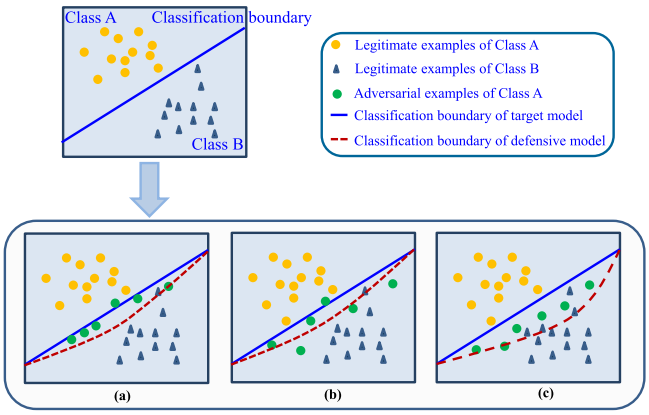


Fig. 4. Relationship between the quality of the adversarial exampels and the defensive model. (a) Almost all the adversarial examples are closer to the classification boundary. (b) Some of the adversarial examples are closer to the classification boundary. (c) Almost all the adversarial examples are far away from the classification boundary.

attack performance from the view of the attack success rate, and the numerical metric on the defensive model illustrates the distance between the adversarial examples and classification boundary. In particular, the distance between the adversarial examples and classification boundary is difficult to measure numerically; therefore, a defensive strategy with adversarial training is implemented for distance measurement. Fig. 4 and Table I illustrate the motivation for evaluating the quality of adversarial examples with a defensive model in a simple way. Fig. 4 is a schematic of class A attacking class B. In detail, adversarial examples generated by class A are misclassified as class B. There are three possible cases of adversarial examples generated on the target model.

1) *Almost All the Adversarial Examples Are Closer to the Classification Boundary [see Fig. 4(a)]:* In this case, the distribution of adversarial examples is relatively concentrated; therefore, through adversarial training, adversarial examples can be correctly classified by moving the classification boundary a small distance. Hence, the classification accuracy of the adversarial examples on the defensive model is significantly increased, while the classification accuracy of legitimate examples does not significantly decrease.
2) *Some of the Adversarial Examples Are Closer to the Classification Boundary [see Fig. 4(b)]:* In this case, the distribution of adversarial examples has a large difference. Some adversarial examples are closer to the classification boundary, and some are far away. Therefore, it is difficult for defensive boundary to fit all adversarial examples with adversarial training, and a large number of adversarial examples are still misclassified on the defensive model.
3) *Almost All the Adversarial Examples Are Far From the Classification Boundary [see Fig. 4(c)]:* In this case, although the adversarial examples are far from the classification boundary, the distribution is relatively concentrated. Therefore, defensive boundary can better fit the adversarial examples with adversarial training, and the adversarial examples can achieve higher classification accuracy on the defensive model. However, the classification boundary of the defensive model will be far from the classification boundary of the target model, which will cause some legitimate examples of the attacked class to be misclassified.

Therefore, the metric values on the target model and defensive model will be jointly analyzed to evaluate the quality of adversarial examples.

C. Experimental Results

In the experiments, three state-of-the-art adversarial attack methods are used for comparisons, including FGSM [17], PGD [18], and C&W [19]. Since the original FGSM, PGD, and C&W methods are proposed for single-modality attacks, for a fair comparison, we extend these methods into multimodal ways. The implementation details are shown in (20)–(22).

Multimodal FGSM Attack:

$$\begin{cases} x'_D = x_D - \varepsilon \operatorname{sgn}(\nabla_{x_D} L(x_D, x_T, t)) \\ x'_T = x_T - \varepsilon \operatorname{sgn}(\nabla_{x_T} L(x_D, X_T, t)). \end{cases} \tag{20}$$

Multimodal PGD Attack:

$$\begin{cases} x_{D,i+1} = \operatorname{clip}_{x_D,\varepsilon}(x_{D,i} + \mu \operatorname{sgn}(\nabla_{x_{D,i}} L(x_{D,i}, x_{T,i}, t))) \\ x_{T,i+1} = \operatorname{clip}_{x_T,\varepsilon}(x_{T,i} + \mu \operatorname{sgn}(\nabla_{x_{T,i}} L(x_{D,i}, x_{T,i}, t))). \end{cases} \tag{21}$$

Multimodal C&W Attack:

$$\min \left\| x_D - x'_D \right\|_2^2 + \left\| x_T - x'_T \right\|_2^2 + cf(x_D, x_T, t). \tag{22}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASSIFICATION ACCURACY OF "VAIHINGEN_20" IN THE ISPRS VAIHINGEN 2D DATASET

Model	Examples	Metric	FGSM	PGD	C&W	MFCANet
Target model	legitimate examples	OA	0.8870	0.8870	0.8870	0.8870
		Kappa	0.8379	0.8379	0.8379	0.8379
		mIoU	0.7391	0.7391	0.7391	0.7391
	Adversarial examples	OA	0.2201	0.2126	0.1293	0.0989
		Kappa	0.0139	-0.0819	-0.0128	-0.1912
		mIoU	0.0891	0.0805	0.0615	0.0678
Defensive model	legitimate examples	OA	0.7382	0.8369	0.8564	0.8334
		Kappa	0.6161	0.6764	0.7995	0.7509
		mIoU	0.4153	0.4940	0.6890	0.6171
	Adversarial examples	OA	0.7762	0.6110	0.6831	0.8325
		Kappa	0.6869	0.5395	0.2706	0.7298
		mIoU	0.4683	0.4258	0.2199	0.6075
	Testing time(h)	–	0.1076	1.3091	0.6078	0.0578

TABLE III
CLASSIFICATION ACCURACY OF "VAIHINGEN_31" IN THE ISPRS VAIHINGEN 2D DATASET

Model	Examples	Metric	FGSM	PGD	C&W	MFCANet
Target model	legitimate examples	OA	0.9074	0.9074	0.9074	0.9074
		Kappa	0.8773	0.8773	0.8773	0.8773
		mIoU	0.7575	0.7575	0.7575	0.7575
	Adversarial examples	OA	0.2609	0.1580	0.1305	0.0961
		Kappa	0.1612	0.0176	0.0133	-0.0687
		mIoU	0.1273	0.0738	0.0745	0.0411
Defensive model	legitimate examples	OA	0.8364	0.8857	0.8825	0.8807
		Kappa	0.7859	0.8558	0.8523	0.8294
		mIoU	0.6019	0.7241	0.7275	0.7012
	Adversarial examples	OA	0.8189	0.7838	0.7609	0.8819
		Kappa	0.7645	0.7355	0.2033	0.8419
		mIoU	0.6184	0.5734	0.1849	0.7070
	Testing time(h)	–	0.3023	3.2195	0.5402	0.2449

TABLE IV
CLASSIFICATION ACCURACY OF "POTSDAM5_15" IN THE ISPRS POTSDAM 2D DATASET

Model	Examples	Metric	FGSM	PGD	C&W	MFCANet
Target model	legitimate examples	OA	0.8472	0.8472	0.8472	0.8472
		Kappa	0.7976	0.7976	0.7976	0.7976
		mIoU	0.6860	0.6860	0.6860	0.6860
	Adversarial examples	OA	0.1688	0.1592	0.1023	0.1428
		Kappa	0.0424	-0.0067	-0.1818	0.0164
		mIoU	0.0839	0.0939	0.0698	0.0613
Defensive model	legitimate examples	OA	0.5005	0.7611	0.8315	0.8253
		Kappa	0.4100	0.6906	0.7780	0.7735
		mIoU	0.358	0.5534	0.6701	0.6663
	Adversarial examples	OA	0.8566	0.4975	0.7555	0.8324
		Kappa	0.8073	0.3508	0.6717	0.7534
		mIoU	0.6762	0.3091	0.5483	0.6223
	Testing time(h)	–	1.8003	27.5865	8.0612	1.2477

In all the compared methods, the hyperparameter ε is fixed to 0.3 for a fair comparison. For the multimodal PGD attack, μ is set as 0.02, and the iteration number is set as 10. For a multimodal C&W attack, the hyperparameter c is set as 10, the learning rate is set as 0.01, and the iteration number is set as 100. Tables II–V show the classification quality metrics with different attack methods. Figs. 5–8 show the classification maps of adversarial examples on the target model and the defensive, respectively.

1) ISPRS Vaihingen 2D Dataset: Tables II and III show the classification accuracy of patch ID numbers "Vaihingen_20" and "Vihingen_31," respectively. On the target model, the OA values of legitimate examples on the two patches are 0.8870 and 0.9074, respectively, but the test accuracies of adversarial examples have a significant drop in all the compared attack methods. The adversarial examples generated by the proposed MFCANet have a strong attack ability, and the OA values are only 0.0989 and 0.0961 on the two patches,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CLASSIFICATION ACCURACY OF "POTSDAM6_13" IN THE ISPRS POTSDAM 2D DATASET

Model	Examples	Metric	FGSM	PGD	C&W	MFCANet
Target model	legitimate examples	OA	0.8939	0.8939	0.8939	0.8939
		Kappa	0.8533	0.8533	0.8533	0.8533
		mIoU	0.7221	0.7221	0.7221	0.7221
	Adversarial examples	OA	0.2756	0.2564	0.1853	0.1925
		Kappa	0.1569	0.1202	-0.0256	-0.0243
		mIoU	0.1326	0.1313	0.0965	0.1012
Defensive model	legitimate examples	OA	0.7148	0.8232	0.8828	0.8866
		Kappa	0.6227	0.7501	0.8407	0.8551
		mIoU	0.4644	0.6064	0.7283	0.7290
	Adversarial examples	OA	0.8456	0.5616	0.8632	0.8886
		Kappa	0.7839	0.4192	0.8109	0.8232
		mIoU	0.5382	0.3542	0.6697	0.6782
Testing time(h)		–	1.8003	27.5865	8.0612	1.2477

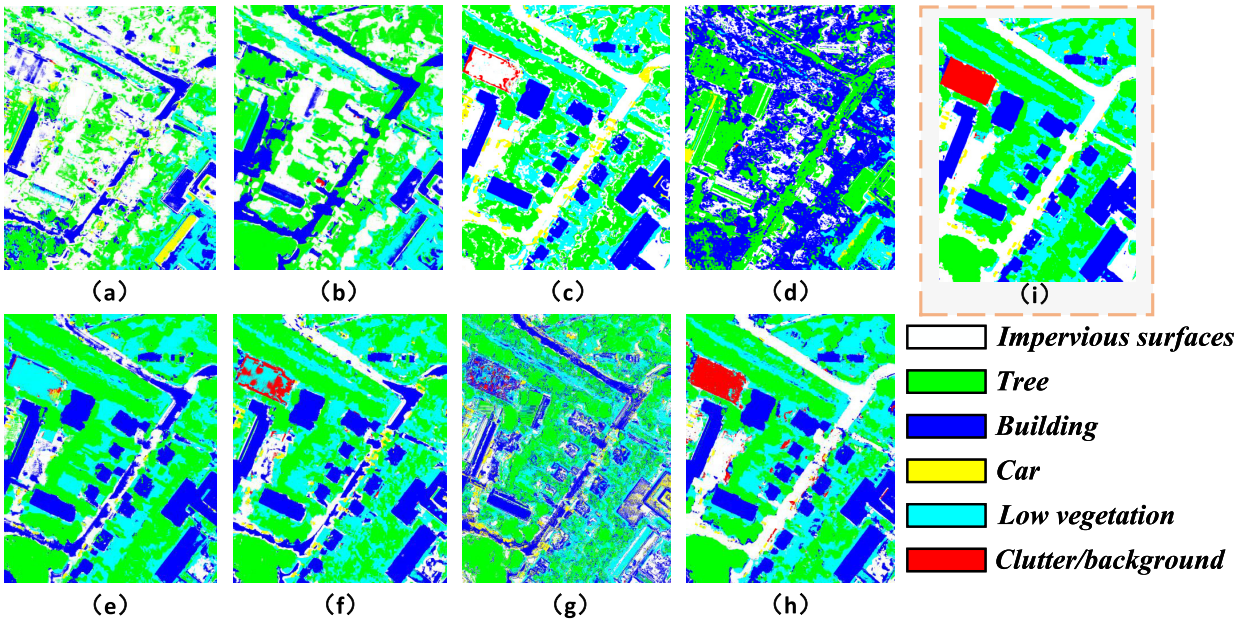


Fig. 5. Classification maps of adversarial examples on patch ID number "Vaihingen_20." (a)–(d) Classification maps of adversarial examples on the target model. (e)–(h) Classification maps of adversarial examples on the defensive model. (a) and (e) Multimodal FGSM. (b) and (f) Multimodal PGD. (c) and (g) Multimodal C&W. (d) and (h) MFCANet (Proposed). (i) Classification map of the legitimate examples on the target model.

respectively. That is, almost all the test adversarial examples are misclassified on the target model. The values of kappa and mIoU also illustrate the effectiveness of the proposed MFCANet.

On the defensive model, we test the OA, kappa, and mIoU values on legitimate examples and adversarial examples. For the multimodal FGSM attack method, the OA, kappa, and mIoU values are low for legitimate examples and high for adversarial examples. In contrast, multimodal PGD and multimodal C&W achieve higher OA, kappa, and mIoU values on legitimate examples, but the defensive model does not perform well against the generated adversarial examples, especially for the multimodal C&W method. According to the analysis in Section IV-B, it can be concluded that the adversarial examples generated by multimodal FGSM are far from the classification boundary, and the adversarial examples generated by multimodal PGD and multimodal C&W methods have large distribution differences. For the proposed method, both the legitimate examples and adversarial examples obtain higher classification accuracy on the defensive model. Therefore, the adversarial examples generated by the proposed MFCANet method are closer to the classification boundary.

In addition, the testing time for generating the adversarial examples and predicting their labels is shown in Tables II and III. The testing time is the time taken to generate adversarial examples for all the test examples. The proposed method has the shortest testing time among the compared methods, while multimodal FGSM is generally considered to have a shorter testing time than multimodal PGD and multimodal C&W. Therefore, the proposed MFCANet method has an advantage in testing time.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

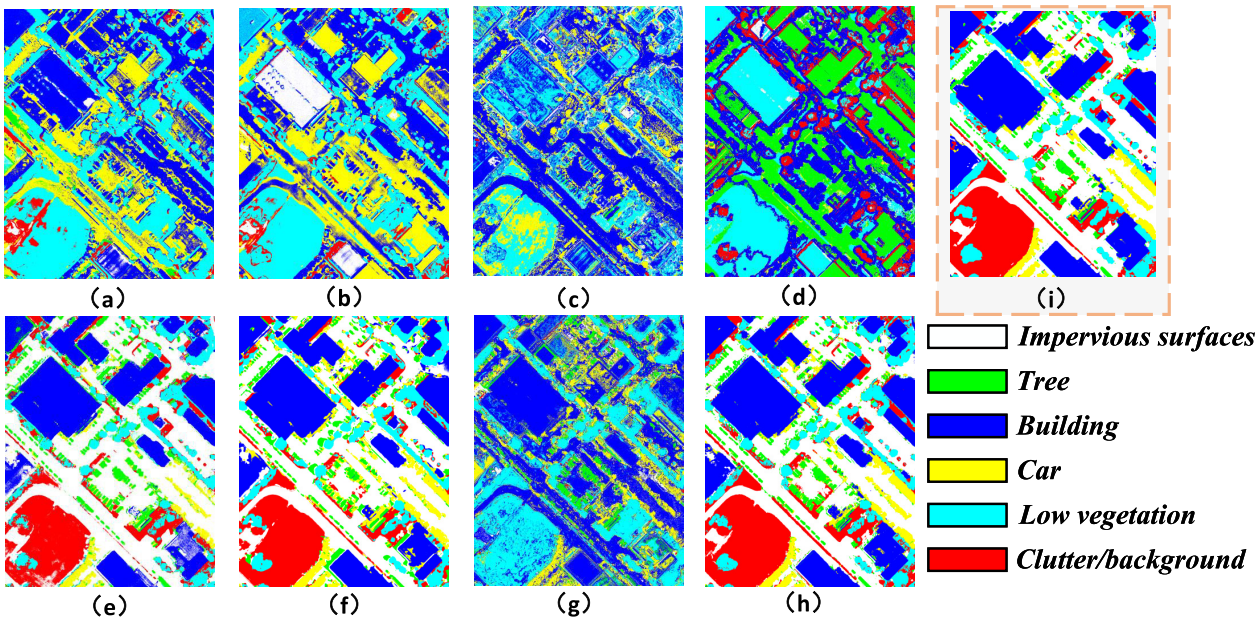


Fig. 6. Classification maps on patch ID number "Vaihingen_31." (a)–(d) Classification maps of adversarial examples on the target model. (e)–(h) Classification maps of adversarial examples on the defensive model. (a) and (e) Multimodal FGSM. (b) and (f) Multimodal PGD. (c) and (g) Multimodal C&W. (d) and (h) MFCANet (Proposed). (i) Classification map of the legitimate examples on the target model.

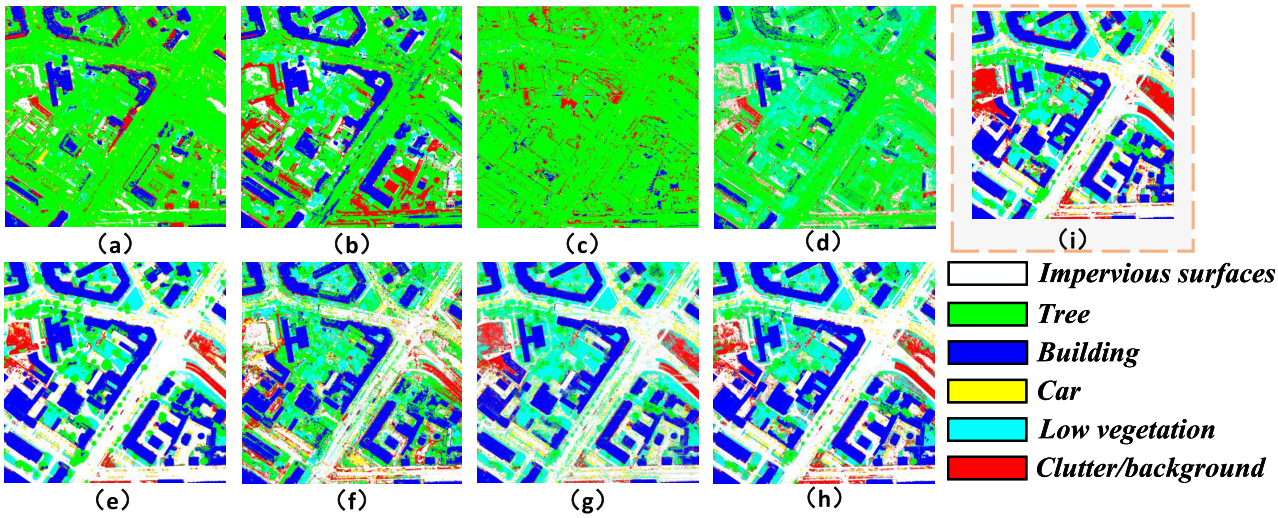


Fig. 7. Classification maps of adversarial examples on patch ID number "Potsdam5_15." (a)–(d) Classification maps of adversarial examples on the target model. (e)–(h) Classification maps of adversarial examples on the defensive model. (a) and (e) Multimodal FGSM. (b) and (f) Multimodal PGD. (c) and (g) Multimodal C&W. (d) and (h) MFCANet (Proposed). (i) Classification map of the legitimate examples on the target model.

Figs. 5 and 6 show the classification maps of adversarial examples on the target model and defensive model, respectively. Compared with the classification maps of legitimate examples on the target model [see Figs. 5(i) and 6(i)], we can clearly see that the target model cannot effectively resist adversarial examples. Taking a closer observation, the results show that the proposed MFCANet method has a higher attack success rate, and most areas are misclassified. On the defensive model, the classification maps obtained by the proposed MFCANet method [see Figs. 5(h) and 6(h)] are closer to the legitimate examples on the target model [see Figs. 5(i) and 6(i)]. Therefore, the proposed MFCANet method outperforms the other methods regarding the attack ability and examples quality.

2) ISPRS Potsdam 2D Dataset: The ISPRS Potsdam 2D dataset has larger and more complex scene information, and it is more difficult to classify. Therefore, we select 10 000 pixels per class to train the target model to ensure higher classification accuracy of legitimate examples. Tables IV and V show the classification accuracies of patch ID numbers "Potsdam5_15" and "Potsdam6_13," respectively. On the target model, the overall accuracies of legitimate examples on the two patches are 0.8472 and 0.8939, respectively. Multimodal C&W outperforms the proposed method

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

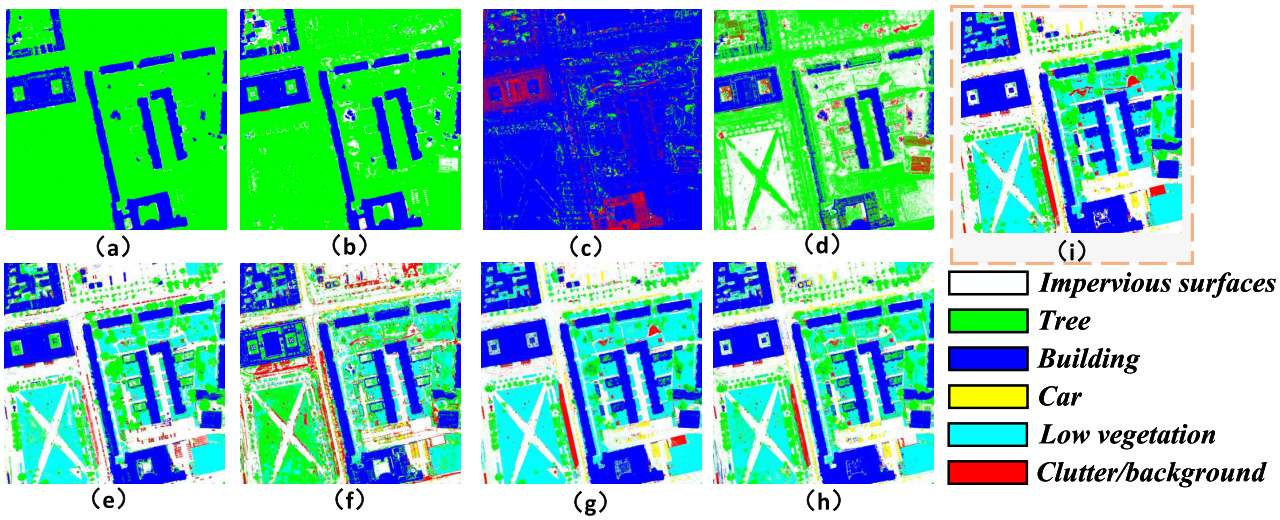


Fig. 8. Classification maps of adversarial examples on patch ID number "Potsdam6_13." (a)–(d) Classification maps of adversarial examples on the target model. (e)–(h) Classification maps of adversarial examples on the defensive model. (a) and (e) Multimodal FGSM. (b) and (f) Multimodal PGD. (c) and (g) Multimodal C&W. (d) and (h) MFCANet (Proposed). (i) Classification map of the legitimate examples on the target model.

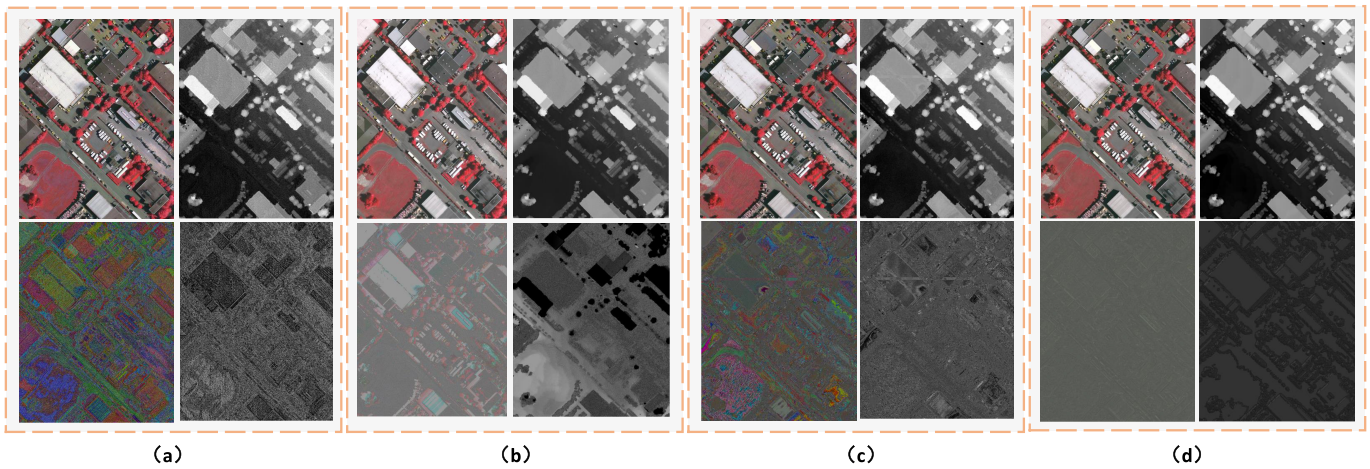


Fig. 9. Generated adversarial examples and corresponding perturbations on patch ID number "Vaihingen_31." (a) Multimodal FGSM. (b) Multimodal PGD. (c) Multimodal C&W. (d) MFCANet (proposed).

regarding the attack performance on the target model, and the attack success is higher than that of the proposed method. However, on the defensive model, the proposed method outperforms multimodal C&W on adversarial examples. As analyzed in Section IV-B, the distribution difference of adversarial examples generated by the proposed method is smaller than that of the multimodal C&W method. For the multimodal FGSM method, the classification accuracy of adversarial examples is still higher than that of legitimate examples on the defensive model, especially as shown in Table IV. These results further illustrate that adversarial examples generated by multimodal FGSM are far away from the classification boundary compared to other methods. In addition, observing the testing time, the advantages of the proposed method in testing large-scale images are more obvious.

Figs. 7 and 8 show the classification maps of adversarial examples on the target model and defensive model, respectively. The classification map of the proposed method has higher misclassification on the target model [see Figs. 7(d) and 8(d)], but the classification map obtained on the defensive model [see Fig. 7(h) and 8(h)] is closer to the original ones [see Figs. 7(i) and 8(i)].

In addition, Fig. 9 shows the visual map of the generated adversarial examples for the patch ID number "Vaihingen_31." The top row shows the generated TOP and DSM adversarial examples, and the bottom row shows the corresponding adversarial perturbations. The darker pixel values represent smaller perturbations. The results show that the multimodal FGSM obtains the largest perturbation on both the TOP and DSM data. The perturbation of multimodal PGD on DSM is better than that of multimodal FGSM, but the perturbation on TOP is still larger. The perturbation generated by the multimodal C&W is concentrated at the edges of the example. The perturbations generated by the proposed method are also concentrated in the edges, but the intensity is smaller than that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

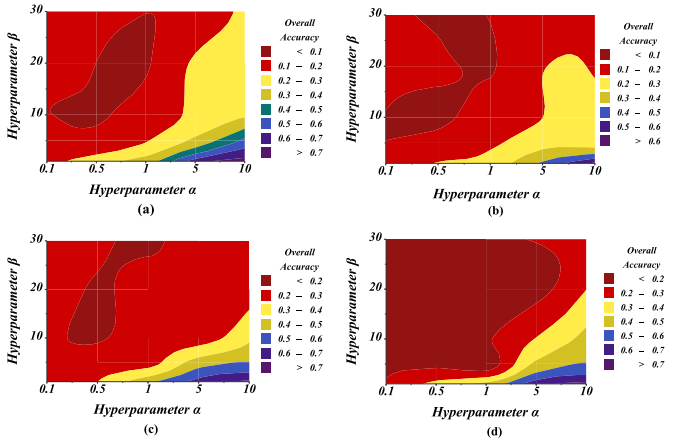


Fig. 10. Study of perceptual weight α and fooling weight β. (a) Vaihingen_20. (b) Vaihingen_31. (c) Potsdam5-15. (d) Potsdam6-13.

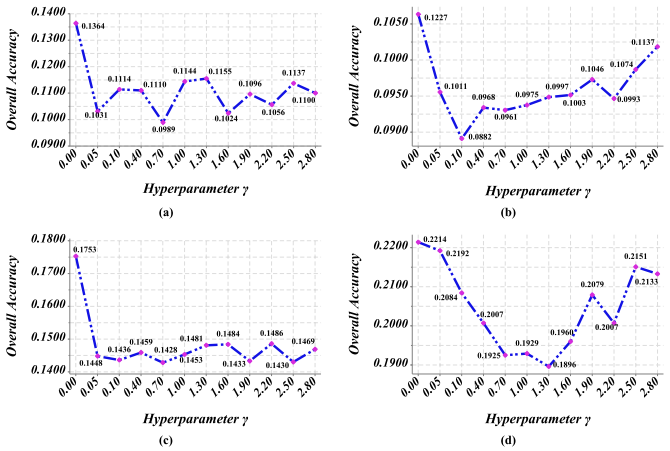


Fig. 11. Study of collaborative weight γ. (a) Vaihingen_20. (b) Vaihingen_31. (c) Potsdam5_15. (d) Potsdam6_13.

of the multimodal C&W. Therefore, the proposed method can achieve stronger attacks with smaller perturbations.

V. Discussion

A. Hyperparameter Study

The three hyperparameters in (12), α, β, and γ, are discussed in this section.

1) Perceptual Weight α and Fooling Weight β: Fig. 10 shows the study of perceptual weight α and fooling weight β. The colors represent the OA value of adversarial examples on the target model. In Fig. 10, the color on the four subfigures has a similar change trend. A larger perceptual weight α and smaller fooling weight β result in a sharp decrease in the attack success rate, and a decrease in perceptual weight α and an increase in fooling weight β can improve the attack ability. However, the reduction in α increases the disturbance intensity in the adversarial examples, which affects the performance of legitimate examples on the defensive model. Therefore, the selection of α and β restricts each other, which need to be balanced in the experiment.

2) Collaborative Weight γ: Fig. 11 shows the study of collaborative weight γ. In Fig. 11, γ ranges from 0 to 2.8. When the value of γ is 0, the collaborative loss term is not used. We can clearly see that the OA value of removing the collaborative loss term is the highest, which means that the attack success rate is the lowest without the collaborative loss term in (12). This observation illustrates the effectiveness of collaborative attacks. In addition, we also observe that when γ is greater than 1.3, the OA value increases and the attack ability decreases, possibly due to the reason that the stronger collaborative constraint reduces the variability of the perturbation on each modality. Therefore, a value of γ between 0.4 and 1.3 can obtain a better attack performance.

B. Study of Collaborative Attack Loss on Perturbation Generation

The collaborative attack loss can effectively improve the multimodal collaborative attack with fewer perturbations. To further illustrate the influence of the collaborative attack loss term [see (12)] on perturbation generation, the perturbations added to the TOP and DSM are analyzed with and without the collaborative attack loss term, respectively. Fig. 12 shows the perturbation change curve on the training set and test set for each epoch. The perturbation value is calculated by the l_∞-norm. $\gamma = 0$ and $\gamma = 0.7$ indicate that there is no collaborative attack loss constraint and that the collaborative weight is 0.7, respectively. The analysis of collaborative weight is discussed in Section V-A in detail. We can notice

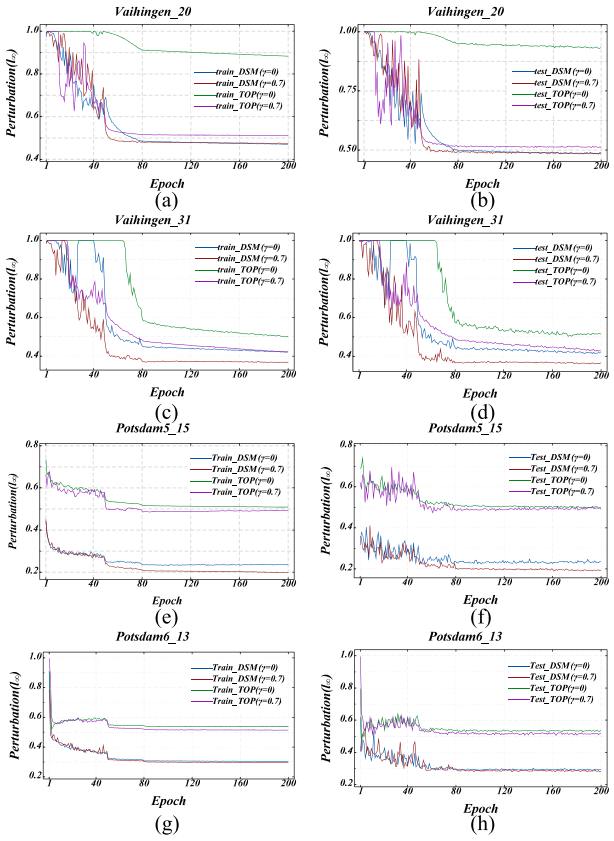


Fig. 12. Perturbation change curve in the training set and test set. (a) and (b) Vaihingen_20. (c) and (d) Vaihingen_31. (e) and (f) Potsdam5_15. (g) and (h) Potsdam6_13.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
OVERALL ACCURACY ON DIFFERENT TARGET MODELS

Patches	Model structure	Target model (OA)		Defensive model (OA)	
		legitimate examples	Adversarial examples	legitimate examples	Adversarial examples
Vaihingen_20	VGG+SC	0.8772	0.0813	0.8307	0.8367
	Resnet+SC	0.8775	0.0924	0.8381	0.8385
	CNN+SKNet	0.8885	0.0797	0.8278	0.8280
Vaihingen_31	VGG+SC	0.8932	0.0977	0.8869	0.8795
	Resnet+SC	0.8970	0.0589	0.8791	0.8523
	CNN+SKNet	0.9114	0.0501	0.8798	0.8706
Potsdam5_15	VGG+SC	0.8413	0.1463	0.8289	0.8377
	Resnet+SC	0.8434	0.1148	0.8235	0.8312
	CNN+SKNet	0.8534	0.0800	0.8121	0.8436
Potsdam6_13	VGG+SC	0.8916	0.1772	0.9055	0.8980
	Resnet+SC	0.9022	0.1541	0.8924	0.8958
	CNN+SKNet	0.8977	0.1533	0.8856	0.8804

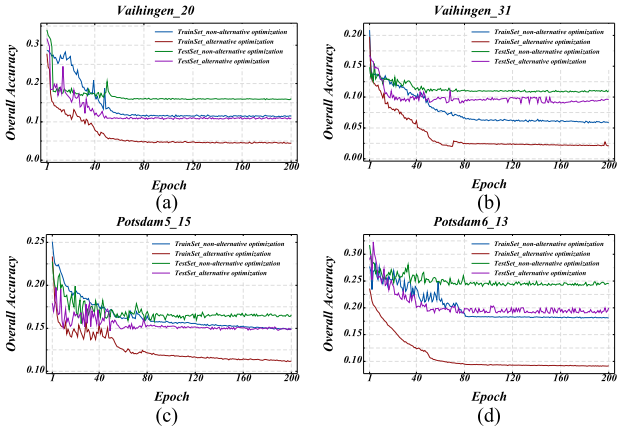


Fig. 13. Classification accuracy by optimizing the objective function in iterative and non-iterative ways. (a) Vaihingen_20. (b) Vaihingen_31. (c) Potsdam5_15. (d) Potsdam6_13.

from Fig. 12 that the perturbations are more inclined to be added on TOP due to its rich color and texture information. For the patch ID numbers "Vaihingen_20" and "Postdam6_13," there is no significant difference in the perturbation added to DSM with $\gamma = 0$ and $\gamma = 0.7$. However, the perturbation added to TOP is obviously reduced with $\gamma = 0.7$, especially for "vaihingen_20." For the patch ID numbers "Vaihingen_31" and "Postdam5_15," the perturbations added on both TOP and DSM are significantly reduced with the collaborative attack loss term. Therefore, the collaborative attack loss can improve the multimodal adversarial attack ability with fewer perturbation by collaborating the perturbation between different modalities.

C. Study of Alternative Optimization

In Section III-B, an alternative optimization method is proposed for optimizing the modality-specific generative loss and modality-specific discriminative loss. To further illustrate the reason for using the alternative optimization strategy, we compare the classification accuracy by optimizing the objective function in an iterative and non-iterative manner, as shown in Fig. 13. It can be seen from the four patches that the iterative optimization method obtains better optimization results on both the training set and test set, and the classification accuracy of the target model is lower than that of the non-iterative optimization method. Therefore, the iterative optimization method can find the optimal solution of the modality-specific generative loss and modality-specific discriminative loss more effectively and achieve a stronger attack effect.

D. Attack Stability Analysis

1) Attack Stability Analysis on Different Target Models: In this section, three state-of-the-art classification models are attacked to verify the attack performance of the proposed method.

1) *VGG + SC:* VGG11 is used to extract the features of each modality, and the modality-specific features are connected by stacked connections.
2) *Resnet + SC:* Resnet18 is used to extract the features of each modality, and the modality-specific features are connected by stacked connections.
3) *CNN + SKNet:* A CNN with a similar structure, as shown in Fig. 3, is used to extract the features of each modality, and the modality-specific features are fused by SKNet [29].

The results are shown in Table VI. In the target model, although different structures are adopted in the feature extraction layer and feature fusion layer, the proposed method has achieved stable attack performance. Compared with the experimental results shown in Tables II–V, the attack performance is even better.

2) Attack Stability Analysis on More Datasets: We select the first five patches per dataset to test the stability of the proposed method, as shown in Fig. 14. Tables VII and VIII show the results of the ISPRS Vaihingen 2D dataset and ISPRS Potsdam 2D dataset. The OA values of legitimate examples and adversarial examples on the target model and defensive model are presented, respectively. The scene information of the ISPRS Vaihingen 2D dataset and ISPRS Potsdam 2D dataset is an urban area, which is more likely to be attacked. Therefore, stronger attack and defense methods are required to protect the target model. As can be seen from Tables VI and VIII, the classification accuracy of the target model is significantly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
STABILITY STUDY: ISPRS VAIHINGEN 2D DATASET

Patch ID number	Image size	Target model(OA)		Defensive model(OA)	
		Legitimate examples	Adversarial examples	Legitimate examples	Adversarial examples
vaihingen_1	1919×2569	0.8569	0.0938	0.8139	0.8103
vaihingen_2	2428×2767	0.8376	0.0852	0.7750	0.7655
vaihingen_3	2006×3007	0.8459	0.1075	0.8189	0.8011
vaihingen_4	1887×2557	0.8590	0.2105	0.8011	0.8333
vaihingen_5	1887×2557	0.8812	0.0591	0.8517	0.8455

TABLE VIII
STABILITY STUDY: ISPRS POTSDAM 2D DATASET

Patch ID number	Image size	Target model(OA)		Defensive model(OA)	
		Legitimate examples	Adversarial examples	Legitimate examples	Adversarial examples
potsdam2-10	6000×6000	0.8442	0.1163	0.8191	0.8308
potsdam2-11	6000×6000	0.8360	0.1196	0.8007	0.7614
potsdam2-12	6000×6000	0.8387	0.0862	0.8313	0.8308
potsdam2-13	6000×6000	0.8346	0.1389	0.8043	0.8020
potsdam2-14	6000×6000	0.8442	0.1319	0.8237	0.8024

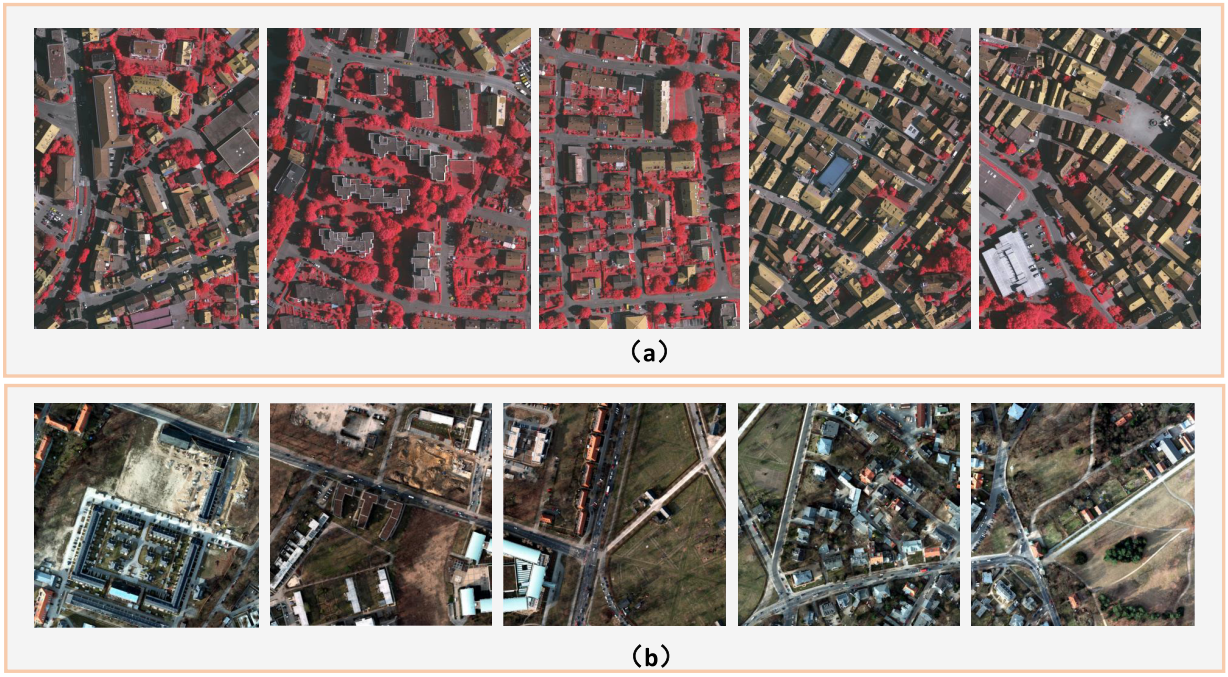


Fig. 14. TOP of the ISPRS Vaihingen 2D dataset and ISPRS Potsdam 2D dataset. (a) ISPRS Vaihingen 2D datasets. (b) ISPRS Potsdam 2D datasets.

reduced under the attack of the proposed MFCANet method. However, in the defensive model, the accuracy of adversarial samples is obviously improved on all the ten patches. Although the five sets of patches have different characteristics, the proposed method shows better attack performance.

VI. CONCLUSION AND FUTURE WORK

In this article, we proposed an MFCANet for attacking multimodal RS classification networks, which consists of two modality-specific generators, two modality-specific discriminators, and a target model. A modality-specific generative loss and a modality-specific discriminative loss are designed for training the proposed MFCANet. The modality-specific generative loss allows the networks to generate multimodal adversarial examples with smaller perturbations, and the adversarial examples between different modalities have collaborative attack ability. The modality-specific discriminative loss is used to improve the quality of the generated multimodal adversarial examples, which makes the generated adversarial examples closer to the real instances. In addition, an alternating optimization strategy is designed to achieve network training. The proposed MFCANet is tested on the ISPRS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.

Vaihingen 2D dataset and ISPRS Potsdam dataset. The results show that the proposed method outperforms the state-of-the-art adversarial attack methods.

Although the proposed method has been proven to be effective and stable, the experimental results show that the classification accuracy of legitimate examples is still affected by adversarial examples on the defensive model. Therefore, improving the attack performance is an issue that needs to be further studied.

REFERENCES

[1] X. Pan, C. Zhang, J. Xu, and J. Zhao, "Simplified object-based deep neural network for very high resolution remote sensing image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 218–237, Nov. 2021.

[2] J. Feng *et al.*, "Generative adversarial networks based on collaborative learning and attention mechanism for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 7, p. 1149, Apr. 2020.

[3] J. Feng *et al.*, "Attention multibranch convolutional neural network for hyperspectral image classification based on adaptive region search," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5054–5070, Jun. 2021.

[4] W. Jiao, L. Wang, and M. F. McCabe, "Multi-sensor remote sensing for drought characterization: Current status, opportunities and a roadmap for the future," *Remote Sens. Environ.*, vol. 256, Apr. 2021, Art. no. 112313.

[5] W. Zhao *et al.*, "Superpixel-based multiple local CNN for panchromatic and multispectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 4141–4156, Jul. 2017.

[6] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[7] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[8] X. Du, X. Zheng, X. Lu, and A. A. Doudkin, "Multisource remote sensing data classification with graph fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10062–10072, Dec. 2021.

[9] C. Szegedy, W. Zaremba, and I. Sutskever, "Intriguing properties of neural networks," in *Proc. 12th Int. Conf. Learn. Represent.*, 2014, pp. 1–10.

[10] G. R. Machado, E. Silva, and R. R. Goldschmidt, "Adversarial machine learning in image classification: A survey toward the defender's perspective," *ACM Comput. Surv.*, vol. 55, no. 1, pp. 1–38, Jan. 2023.

[11] X. Yuan, P. He, Q. Zhu, and X. Li, "Adversarial examples: Attacks and defenses for deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2805–2824, Sep. 2019.

[12] Y. Duan, X. Zhou, J. Zou, J. Qiu, J. Zhang, and Z. Pan, "Mask-guided noise restriction adversarial attacks for image classification," *Comput. Secur.*, vol. 100, Jan. 2021, Art. no. 102111.

[13] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.

[14] L. Chen, Z. Xu, Q. Li, J. Peng, S. Wang, and H. Li, "An empirical study of adversarial examples on remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7419–7433, Sep. 2021.

[15] H. Li *et al.*, "Adversarial examples for CNN-based SAR image classification: An experience study," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1333–1347, 2021.

[16] Y. Xu, B. Du, and L. Zhang, "Self-attention context network: Addressing the threat of adversarial attacks for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8671–8685, 2021.

[17] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. 13th Int. Conf. Learn. Represent.*, 2015, pp. 1–11.

[18] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Proc. 5th Int. Conf. Learn. Represent.*, 2017, pp. 1–14.

[19] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

[20] Q. Liang, Q. Li, and S. Yang, "LP-GAN: Learning perturbations based on generative adversarial networks for point cloud adversarial attacks," *Image Vis. Comput.*, vol. 120, Apr. 2022, Art. no. 104370.

[21] F. Liu, M. Xu, G. Li, J. Pei, L. Shi, and R. Zhao, "Adversarial symmetric GANs: Bridging adversarial samples and adversarial networks," *Neural Netw.*, vol. 133, pp. 148–156, Jan. 2021.

[22] C. W. Xiao, B. Li, and J. Y. Zhu, "Generating adversarial examples with adversarial networks," 2019, *arXiv:1801.02610*.

[23] S. Jandial, P. Mangla, S. Varshney, and V. Balasubramanian, "AdvGAN++: Harnessing latent layers for adversary generation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2045–2048.

[24] X. Liu and C.-J. Hsieh, "Rob-GAN: Generator, discriminator, and adversarial attacker," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11226–11235.

[25] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[26] C. Du, C. Huo, L. Zhang, B. Chen, and Y. Yuan, "Fast C&W: A fast adversarial attack algorithm to fool SAR target recognition with deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[27] C. Shi, Y. Dang, L. Fang, Z. Lv, and M. Zhao, "Hyperspectral image classification with adversarial attack," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] A. Sajeeda and B. M. M. Hossain, "Exploring generative adversarial networks and adversarial training," *Int. J. Cognit. Comput. Eng.*, vol. 3, pp. 78–89, Jun. 2022.

[29] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.



Cheng Shi (Member, IEEE) received the B.S. degree from the Xi'an University of Architecture and Technology, Xi'an, Shaanxi, China, in 2009, and the M.S. and Ph.D. degrees from Xidian University, Xi'an, in 2012 and 2016, respectively.

She has worked as a Post-Doctoral Research Fellow with the University of Macau, Macau, China, from 2016 to 2018. She is currently working as an Associate Professor with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an. Her research interests include deep learning, remote sensing image classification, and pattern recognition.



Yenan Dang received the B.S. degree from Xi'an Petroleum University, Xi'an, Shaanxi, China, in 2020. She is currently pursuing the M.S. degree with the Xi'an University of Technology, Xi'an.

Her main research directions include deep learning and remote sensing image classification.



Li Fang (Member, IEEE) received the bachelor's degree in land resources management and the master's degree in physical geography from the China University of Geosciences, Wuhan, China, in 2007 and 2009, respectively, and the master's degree in geoinformatics and the Ph.D. degree in photogrammetry and remote sensing from the Technical University of Munich, Munich, Germany, in 2011 and 2017, respectively.

His main research interests include image processing and analysis, pattern recognition, and related remote sensing/environmental and industrial applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.



Minghua Zhao received the Ph.D. degree in computer science from Sichuan University, Chengdu, China, in 2006.

After that, she joined the Xi'an University of Technology, Xi'an, China, where she is currently a Professor with the School of Computer Science and Engineering. Her research interests include image processing, pattern recognition, and computer vision.



Zhiyong Lv (Member, IEEE) received the M.S. and Ph.D. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2008 and 2014, respectively.

He has worked as an Engineer of surveying with The First Institute of Photogrammetry and Remote Sensing, Xi'an, China, from 2008 to 2011. He is currently working as an Associate Professor with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an. His research interests include multi-hyperspectral and high-resolution remotely sensed image processing, spatial feature extraction, neural networks, pattern recognition, deep learning, and remote sensing applications.



Qiguang Miao (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in computer science from Xidian University, Xi'an, China, in 2004 and 2005, respectively.

He is currently a Professor with the School of Computer Science and Technology, Xidian University. His research interests include intelligent image processing and machine learning.



Chi-Man Pun (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in software engineering from the University of Macau, Macau, China, in 1995 and 1998, respectively, and the Ph.D. degree in computer science and engineering from The Chinese University of Hong Kong, Hong Kong, in 2002.

He was the Head of the Department of Computer and Information Science from 2014 to 2019, and is currently an Associate Professor with the Faculty of Science and Technology, University of Macau. He has investigated several funded research projects as a PI and has authored/coauthored more than 150 refereed scientific articles in international journals (including the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, and IEEE TRANSACTIONS ON IMAGE PROCESSING) and conferences (including AAAI, ICDE, SIGGRAPH, and VR). His research interests include image processing and pattern recognition, information security and forensic for multimedia, and adversarial machine learning.

Dr. Pun is a Professional Member of ACM. He has served as the General Chair for the 10th and 11th International Conference on Computer Graphics, Imaging and Visualization (CGIV2013 and CGIV2014) and the 13th IEEE International Conference on e-Business Engineering (ICEBE2016), the General Co-Chair for the IEEE International Conference on Visual Communications and Image Processing (VCIP2020), and the program chair/reviewer for several other international conferences and journals.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:57:33 UTC from IEEE Xplore. Restrictions apply.